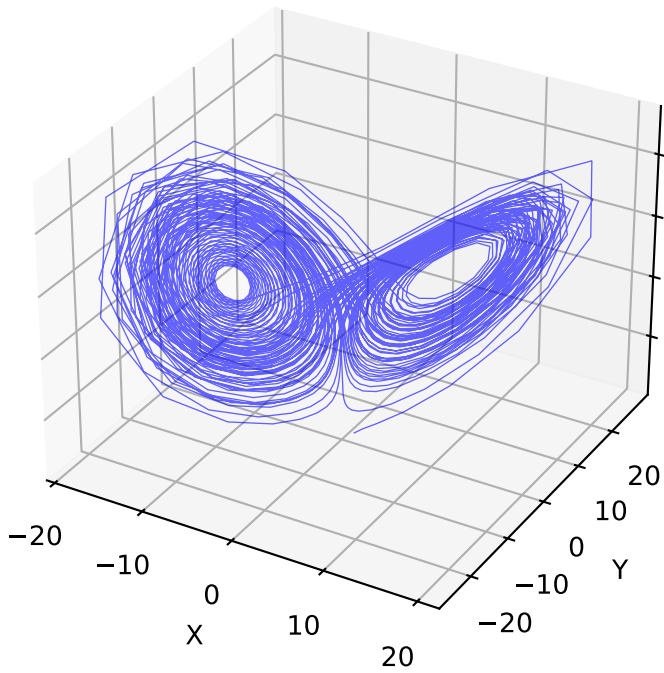
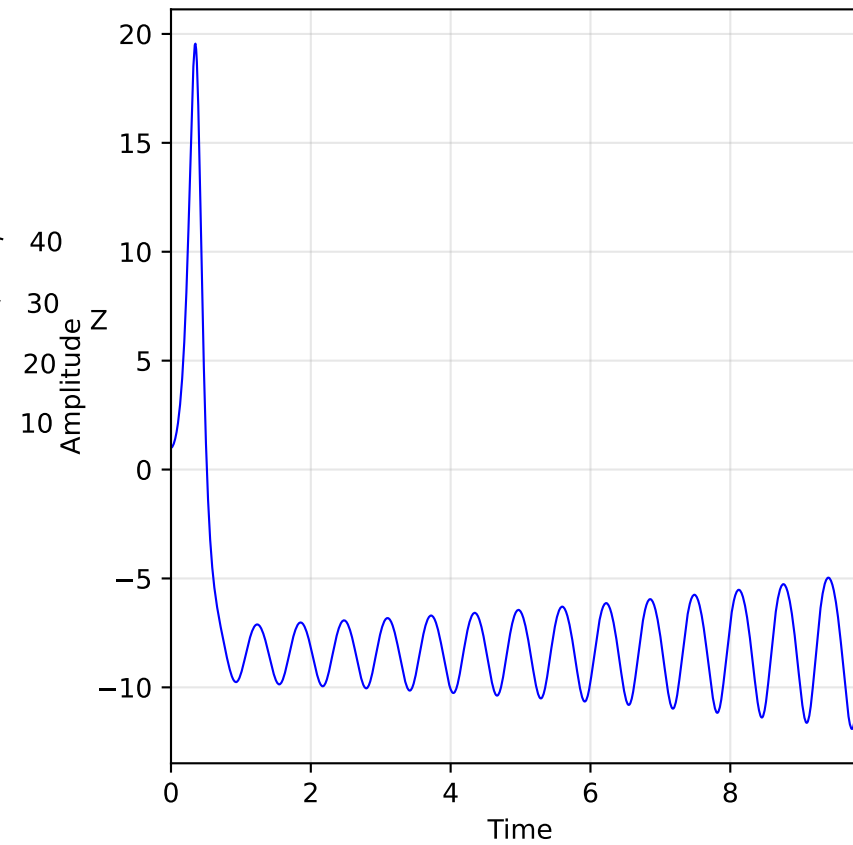


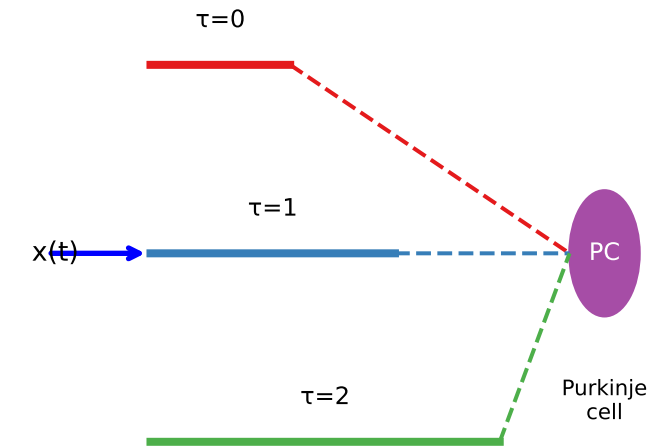
(a) Original Lorenz Attractor
(Ground truth dynamics)



(b) Single-channel observation
("slow wave" signal)

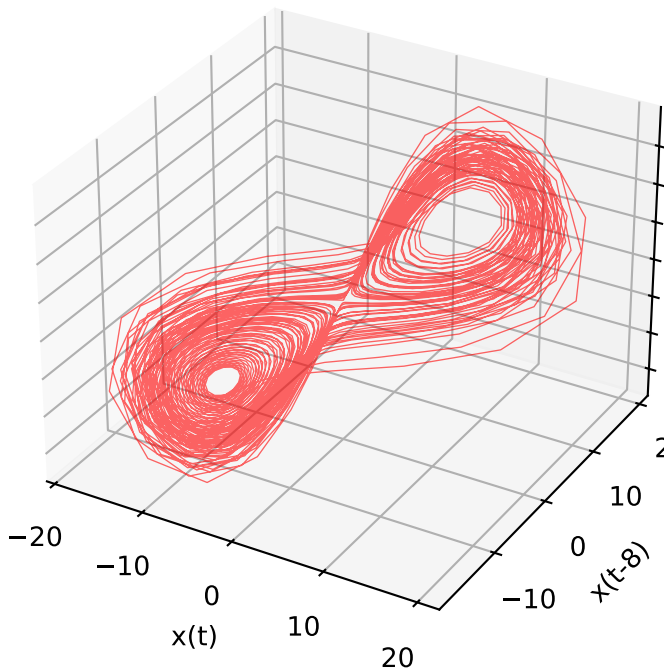


(c) Cerebellar parallel fiber delays

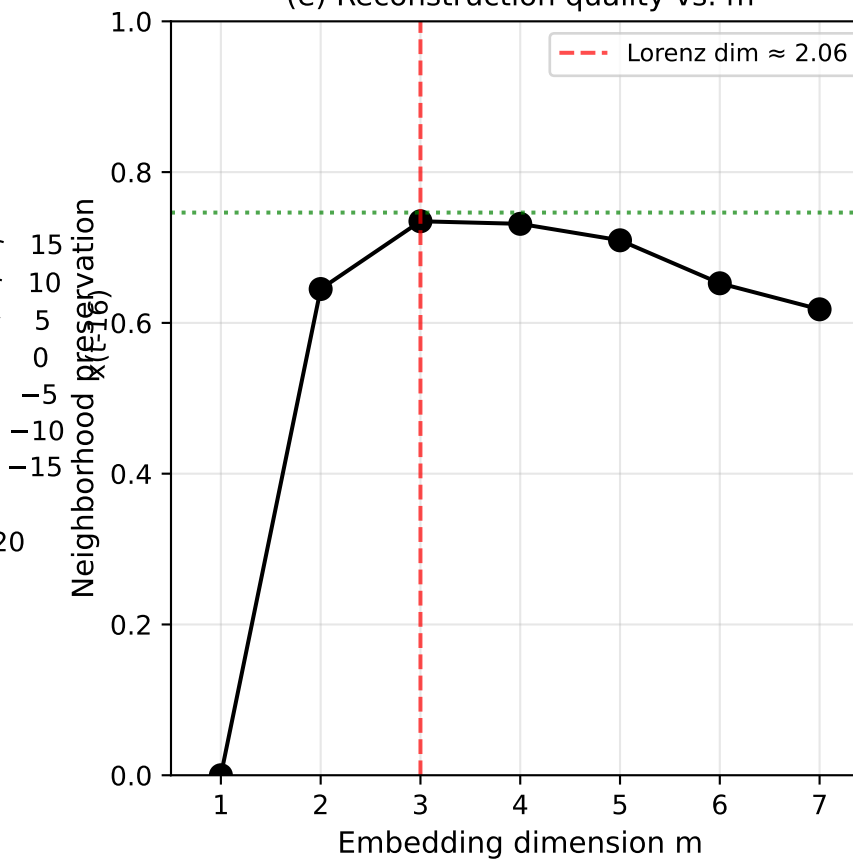


Different fiber lengths $\rightarrow$ delay embedding

(d) Takens reconstruction
($\tau=8$, $m=3$)



(e) Reconstruction quality vs. m



KEY INSIGHT

Takens' theorem: A single scalar observable, properly delay-embedded, recovers the full attractor topology.

Cerebellar parallel fibers have graded lengths
 $\rightarrow$ systematic conduction delays
 $\rightarrow$ natural implementation of delay embedding

CLINICAL RELEVANCE FOR MND:

- Slow cortical oscillations (1-10 Hz) carry high-dimensional coordination signals
- Cerebellum reconstructs full dynamics from delay-tapped slow input
- Loss of slow-wave structure $\rightarrow$ loss of reconstructible coordination manifold
- Coordination fails BEFORE cell death because the manifold becomes unreadable